

1.1: Basic Concepts and Taylor's Theorem

- 1) Show that $|x^2 - 4| < \epsilon$ when $0 < |x - 2| < \epsilon(5 + \epsilon)^{-1}$ and prove $\lim_{x \rightarrow 2} x^2 = 4$ by using these inequalities.

Solution:

From the given inequalities, we can deduce the following:

$$\begin{aligned} 0 < |x - 2| < \epsilon(5 + \epsilon)^{-1} &\implies |x - 2| < \epsilon \\ \implies |x + 2| = |x - 2 + 4| & \\ &\leq |x - 2| + |4| \\ &= |x - 2| + 4 \\ &< 4 + \epsilon \end{aligned}$$

Thus, from this inequality we can conclude the following as desired:

$$\begin{aligned} |x^2 - 4| &= |x + 2||x - 2| \\ &< \frac{(4 + \epsilon)\epsilon}{(5 + \epsilon)} \\ &< \epsilon \end{aligned}$$

□

- 31) Determine the Taylor polynomial of degree 2 for $f(x) = e^{(\cos x)}$ expanded about the point π .

Solution:

Given $f(x) = e^{(\cos x)}$, to determine the Taylor Polynomial of degree 2, it suffices to differentiate the function first:

$$\begin{aligned} f(x) &= e^{(\cos x)} \\ f'(x) &= (-\sin(x))e^{(\cos x)} \\ f''(x) &= (\sin(x))^2 e^{(\cos x)} - \cos(x)e^{(\cos x)} \end{aligned}$$

Then, evaluating each differentials about the point the $x = \pi$ to yield the following:

$$\begin{aligned} f(\pi) &= e^{-1} \\ f'(\pi) &= (-\sin(\pi))e^{(\cos \pi)} = 0 \\ f''(\pi) &= (\sin(\pi))^2 e^{(\cos \pi)} - \cos(\pi)e^{(\cos \pi)} = e^{-1} \end{aligned}$$

Hence, the Taylor polynomial expansion of the given function yields

$$e^{(\cos(x))} = e^{-1} + \frac{1}{2}(x - \pi)^2 e^{-1}$$

□

1.2: Orders of Convergence and Additional Basic Concepts

- 2) Let a sequence x_n be defined inductively by $x_{n+1} = F(x_n)$. Suppose that $x_n \rightarrow x$ as $n \rightarrow \infty$ and $F'(x) = 0$. Show that

$$x_{n+2} - x_{n+1} = \mathcal{O}(x_{n+1} - x_n).$$

Hint: Use the Mean-Value Theorem and assume that F is a continuously differentiable function.

Solution:

Let F be a continuously differentiable function. Suppose that $x_n \rightarrow x$ as $n \rightarrow \infty$ and $F'(x) = 0$, then for some ξ such that $x_n < \xi < x_{n+1}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{x_{n+2} - x_{n+1}}{x_{n+1} - x_n} &= \lim_{n \rightarrow \infty} \frac{F(x_{n+1}) - F(x_n)}{x_{n+1} - x_n} \\ x_n \rightarrow x &\implies x_{n+1} \rightarrow x \\ &\implies \xi \rightarrow x \\ \implies \lim_{n \rightarrow \infty} F'(\xi) &= \lim_{n \rightarrow \infty} \frac{F(x_{n+1}) - F(x_n)}{x_{n+1} - x_n} \\ &= 0 \end{aligned}$$

□

- 22) Prove that if $0 < \theta < 1$, then $\frac{(1+a\theta^n)}{1+a\theta^{n-1}}$ converges to 1 linearly.

Solution:

Let $0 < \theta < 1$, then to show the convergence rewrite $x_n = \frac{(1+a\theta^n)}{1+a\theta^{n-1}}$. By the definition of convergence, we want to find c and N such that $0 < c < 1$ and $n \geq N$ implies

$$\begin{aligned} |x_{n+1} - 1| &< c|x_n - 1| \\ \implies \frac{x_{n+1} - 1}{x_n - 1} &= \frac{(1+a\theta^{n+1})/(1+a\theta^n) - 1}{(1+a\theta^n)/(1+a\theta^{n-1}) - 1} \\ &= \frac{(1+a\theta^{n+1} - 1 - a\theta^n)/(1+a\theta^n)}{(1+a\theta^n - 1 - a\theta^{n-1})/(1+a\theta^{n-1})} \\ &= \theta \left(\frac{1+a\theta^{n-1}}{1+a\theta^n} \right) \end{aligned}$$

Since $0 < \theta < 1$, then the term $\theta \left(\frac{1+a\theta^{n-1}}{1+a\theta^n} \right)$ converges to 1. Choose N such that for $n \geq N$

$$\theta \left(\frac{1+a\theta^{n-1}}{1+a\theta^n} \right) < \frac{1}{2}(\theta + 1).$$

Then we can use $c = \frac{1}{2}(\theta + 1) < 1$ for $n \geq N$ to show that $\frac{(1+a\theta^n)}{1+a\theta^{n-1}}$ converges to 1 linearly.

□